GE Drive Systems

INSTRUCTIONS

531X157APCA_G1
ANALOG PROCESS INTERFACE CARD

Renewal Part

IMPORTANT
INFORMATION

CAUTION: To ensure proper operation of the motor drive controller, these instructions must be followed for proper set-up and installation of this replacement card.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

SAFETY PRECAUTIONS

WARNING - ELECTRICALSHOCK AND BURN HAZARD: Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

NOTE: *Read all WARNINGS, CAUTIONS and NOTES in the applicable Drive Instruction Book (GEK) prior to removing or installing any card.*

CARD REPLACEMENT PROCEDURE

1. Carefully disconnect all cables.
2. Remove the four screws securing the Analog Process Interface card (Figure 1) to the standoffs on the Main Control card and remove card.
3. Set all jumpers on the replacement card as shown in TABLE 2 for the card version being replaced.
4. Install the new card by reinserting and tightening the four screws removed in step 2.
5. Reconnect all cables verifying that they are properly seated at both ends.
6. Set all potentiometers on the replacement card as shown in TABLE 2 for the card version being replaced and fine tune as required per instructions in the Adjustments Section of the applicable Drive instruction Book.

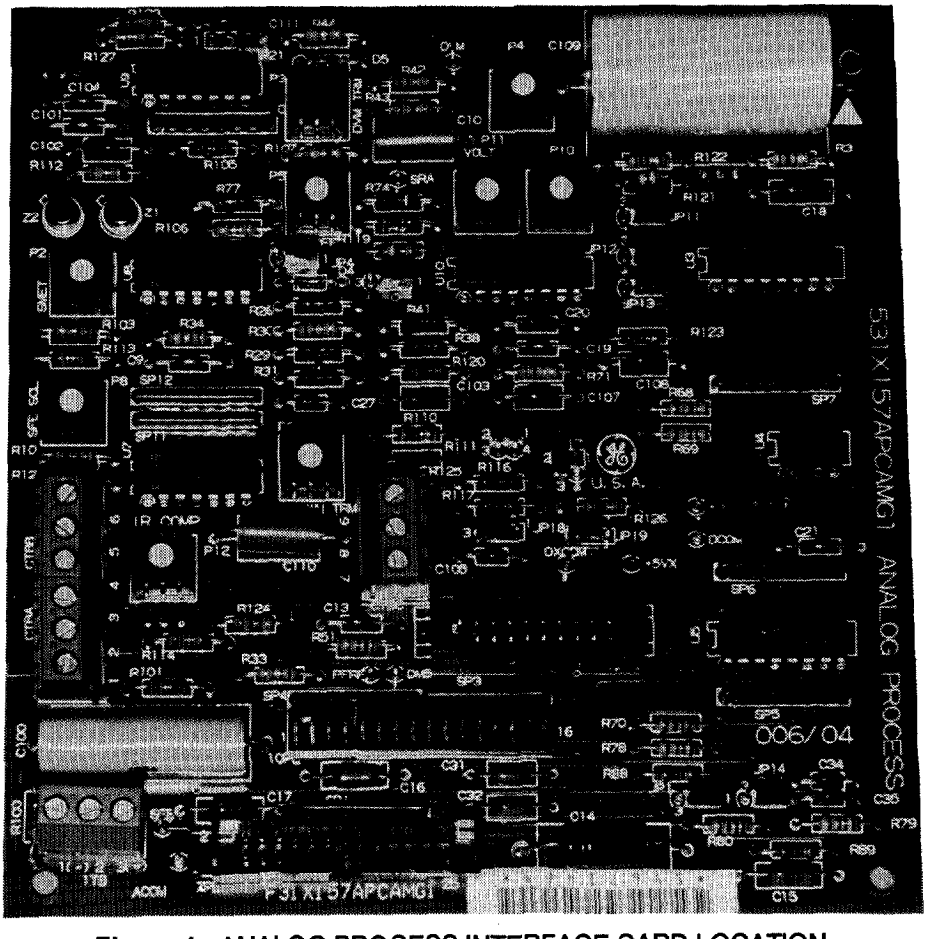


Figure 1. ANALOG PROCESS INTERFACE CARD LOCATION

JUMPER CONFIGURATION

This circuit card is a functional replacement for the cards specified in TABLE 2 provided in this instruction. In order to incorporate design enhancements while continuing to provide compatibilty with a variety of previous designs, changes in card layout (including the relocation, addition or deletion of certain jumpers) may have been required. Jumper locations are shown in Figure 2. Using TABLE 2 of this instruction as a guide, configure the jumpers of the new card as follows:

1. Locate the card numbers for the old and replacement cards.
   -- The part number for the card is printed on a white label that is affixed to the card and begins with the number "531X". This is NOT the number "F531X___ which is silkscreened directly on the circuit board.
2. Cross reference between the card number and configuration jumpers per TABLE 2 for the version card being replaced and set the jumpers per the applicable card interchange listed.
   -- See TABLE 1 for a functional description of the jumpers.
3. Record any jumper setting changes on the Jumper Listing provided with the drive (located in the drive door pocket with the Parameter List).

IMPORTANT: Certain versions of cards may have a part number suffix such as "S" or "H", instead of "G". These cards are functionally identical to the "G" version. When ordering spare or replacement cards, be sure to use the part number on the card or on the Renewal Parts List provided with the drive.

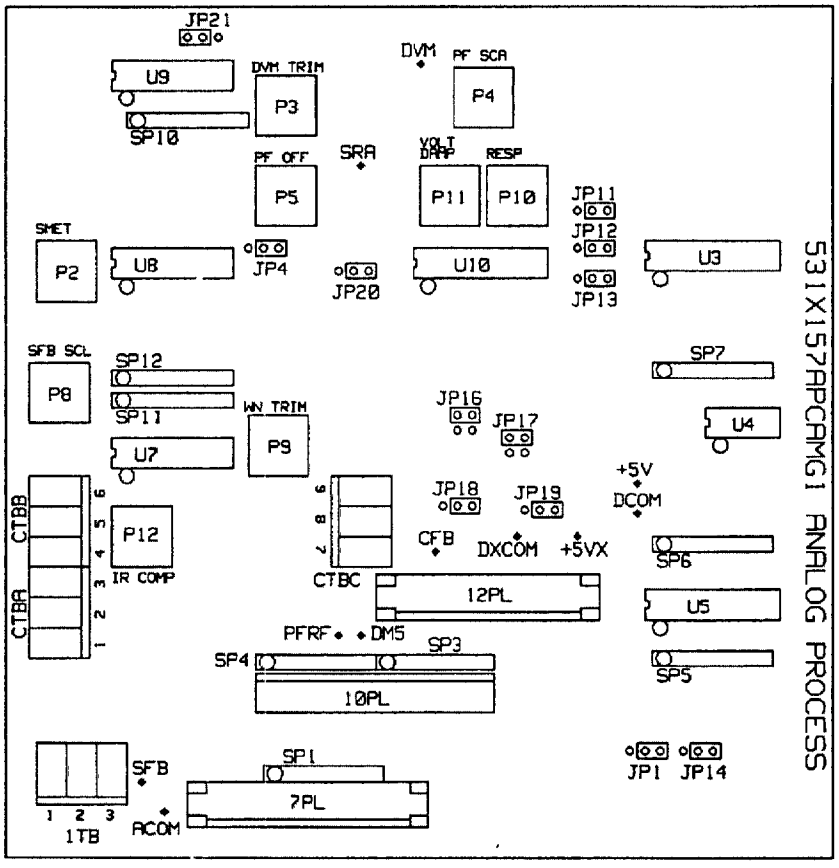


Figure 2. ANALOG PROCESS INTERFACE CARD LAYOUT DIAGRAM

TABLE 1: JUMPER DESCRIPTION - 531X157APCA_G1 CARDS

Location	Jumper or Pot	Number	Nomenclature	Description
Analog Process Interface Card	Pot	P2	SMET	Adjusts the analog output signal at 3TB20 of the speed metering circuit for 10 volts at maximum motor speed. Maximum current load on this signal is 2mA.
		P3	DVM TRIM	With DVM test point connected to +5V and optional Programmer set to DVM operation, adjust P3 until the Programmer displays 5.00.
		P4	PF SCALE	Adjustment for proper running speed. Typically adjusted for 10% maximum tach voltage at 10% reference command. Adjust only after P5 on the Analog Process Interface card and P4 on the Main Control card have been adjusted and customer reference zero is set.
		P5	PF OFF	Zero speed offset adjustment. Typically adjusted for no motor movement when zero speed is commanded. Adjust only after P4 on the Main Control card and customer reference zero have been adjusted.
		P8	SFB SCALE	SFB scaling adjustment. Typically adjusted for 0.4 volts from SFB to A COM at 10% speed command. Final adjustment is for 4 volts from SFB to ACOM at 100% speed. Adjust only after P4 PF SCALE has been adjusted.
		P9	WNTRIM	Adjustment for resonant frequency point of the sharp notch filter in speed regulator. JP16 must be in 1-2 position if used.
		P10	RESPONSE	Speed loop response adjustment. Sets optimum response for best drive performance; adjust per application (clockwise increases gain).
		P11	VOLT DAMP	Adjusts damping effect on speed loop response. Adjust per application (clockwise rotation iscreases time lag between reference and speed feedback).
		P12	IR COMP	IR Drop Compensation for speed regulator. If used, CW increases compounding and full CCW provides zero compounding.
	Jumper	JP1		RXD Mode Select (Analog Tach Only) **1-2:** Enables RS422 reception on the serial link data line (Default position). **2-3:** Enables RS232C reception.
		JP4		SMET Type Select **1-2:** Unipolar speed meter driver mode (Default Pos.) **2-3:** Bipolar meter driver mode (DC tach only).

TABLE 1: JUMPER DESCRIPTION - 531X157APCA_G1 CARDS (cont.)

Location	Jumper or Pot	Number	Nomenclature	Description
Analog Process Interface Card (continued)	Jumper	JP11		TXD Mode Select **1-2:** 12PL-9, 12PL-12 RS422 transmitted data mode (Default position). **2-3:** 12PL-9 RS232 transmitted data mode.
		JP12		DTR Mode Select **1-2:** 12PL-8, 12PL-13 RS422 DTR handshake mode (Default position). **2-3:** 12PL-8 RS232 DTR handshake mode.
		JP13		DTR RS232 Logic Polarity **1-2:** 12PL-8 high=drive ready to receive (Default position). **2-3:** 12PL-8 low=drive ready to receive.
		JP14		CTS Mode Select **1-2:** For RS422 CTS handshaking (Default position). **2-3:** For RS232 CTS handshaking. **NOTE:** *For Series 6 interfaces using the CCM card, set JP1 and JP11 to 1-2 and JP12 and JP13 to 2-3. 12PL-7 should be tied to +5V.*
		JP16		Notch Filter Select **1-2:** Notch filter enabled (Default position). **2-3:** Notch filter disabled and CTBB-6 grounded. **1-4:** Notch filter disabled and CTBB-6 open circuit. **3-4:** Never
		JP17		High Proportional Gain Resistor Select (475 K) **1-2:** High proportional gain resistor selected. **2-3:** High proportional gain resistor not selected (default position). **1-4:** All proportional gain resistors shorted.
		JP18		Medium Proportional Gain Resistor Select (221 K) **1-2:** Medium proportional gain resistor selected. **2-3:** Medium proportional gain resistor not selected (default position).
		JP19		Low Proportional Gain Resistor Select (121 K) **1-2:** Low proportional gain resistor selected. **2-3:** Low proportional gain resistor not selected (default position).
		JP20		Tach Lead Enable Jumper **1-2:** Additional 10ms* tach lead not selected (Default position). **2-3:** Additional 10ms* tach lead selected. **20ms for AJ version cards.*

TABLE 1: JUMPER DESCRIPTION - 531X157APCA_G1 CARDS (cont.)

Location	Jumper or Pot	Number	Nomenclature	Description
Analog Process Interface Card (cont.)	Jumper	JP21		Current Reference Input Enable **1-2:** Input disabled (Default position). **2-3:** Input enabled with 10 volts at 10PL-10 equal to 100% current signal.

"WIRE JUMPERS"

Card	Connection Points	Description
Analog Process Interface card	SFB to DM5	Feeds SFB through a lead network to the summing junction. The strength of the signal is adjustable through P11. The effect of the signal is dampened output response with increasing signal.
Analog Process Interface Card to Main Control Card	CFB to CFB*	(Optional) Provides signal for IR compensation adjustment.
Analog Process Interface Card	SRA to JP18, pin 2	(Optional) To be used when above mentioned IR compensation is used. Clamps integral gain to a specific value. IR comensation now provides additional signal to overcome loss of integral gain component. **NOTE:** *No other JP18 jumpers allowed when this wire jumper is used.*

**CFB on Analog Process Interface card to CFB on Main Control Card.*

TABLE 2: INTERCHANGE MATRIX/INSTRUCTIONS 531X157APCA_G1 CARDS

Replacing AF Version card with AG Version:

1. Transfer all terminal board components.
2. Set jumpers in the following positions (in reference to AF Version card):

JP1	Same	JP14	Same
JP4	Same	JP16	Same
JP11	Same	JP17	Same
JP12	Same	JP18	Same
JP13	Same		

3. Connect "Wire Jumpers" as follows:

Old Postion	New Position
VFBB to DM5	SFB to DM5

4. Set potentiometer positions visually (in reference to AF Version card) in the following positions for preliminary set-up:

P2	Same	P9	Same
P3	Factory Set	P10	Same
P4	Same	P11	Start Full CCW,
P5	Same		Increase as Needed.
P8	Same	P12	Full CW

Replacing AF Version card with AJ Version:

1. Transfer all terminal board components.
2. Set jumpers in the following positions (in reference to AF Version card):

JP1	Same	JP16	Same
JP4	Same	JP17	Same
JP11	Same	JP18	Same
JP12	Same	JP19	2-3:
JP13	Same	JP20	1-2:
JP14	Same	JP21	1-2:

3. Connect "Wire Jumpers" as follows:

Old Postion	New Position
VFBB to DM5	SFB to DM5

4. Set potentiometer positions visually (in reference to AF Version card) in the following positions for preliminary set-up:

P2	Same	P9	Same
P3	Factory Set	P10	Same
P4	Same	P11	Start Full CCW,
P5	Same		Increase as Needed.
P8	Same	P12	Full CW

Replacing AG Version card with AJ Version:

1. Transfer all terminal board components.
2. Set jumpers in the following positions (in reference to AG Version card):

JP1	Same	JP16	Same
JP4	Same	JP17	Same
JP11	Same	JP18	Same
JP12	Same	JP19	2-3:
JP13	Same	JP20	1-2:
JP14	Same	JP21	1-2:

3. Connect "Wire Jumpers" as follows:

Old Postion	New Position
SFB to DM5	Same
CFB to CFB*	Same (if present)
SRA to JP18, pin 2	Same (if present)

4. Set potentiometer positions visually (in reference to AG Version card) in the following positions for preliminary set-up:

P2	Same	P9	Same
P3	Factory Set	P10	Same
P4	Same	P11	Same
P5	Same	P12	Same
P8	Same		

Replacing AF Version card with AK *or* AM Versions:

1. Transfer all terminal board components.
2. Set jumpers in the following positions (in reference to AF Version card):

JP1	Same	JP16	Same
JP4	Same	JP17	Same
JP11	Same	JP18	Same
JP12	Same	JP19	2-3:
JP13	Same	JP20	1-2:
JP14	Same	JP21	1-2:

3. Connect "Wire Jumpers" as follows:

Old Postion	New Position
VFBB to DM5	SFB to DM5

4. Set potentiometer positions visually (in reference to AF Version card) in the following positions for preliminary set-up:

P2	Same	P9	Same
P3	Factory Set	P10	Same
P4	Same	P11	Start Full CCW,
P5	Same		Increase as Needed.
P8	Same	P12	Full CW

**CFB on Analog Process Interface card to CFB on Main Control Card.*

TABLE 2: INTERCHANGE MATRIX/INSTRUCTIONS 531X157APCA_G1 CARDS (continued)

Replacing AG Version card with AK *or* AM Versions:

1. Transfer all terminal board components.
2. Set jumpers in the following positions (in reference to AG Version card):

JP1	Same	JP16	Same
JP4	Same	JP17	Same
JP11	Same	JP18	Same
JP12	Same	JP19	2-3:
JP13	Same	JP20	1-2:
JP14	Same	JP21	1-2:

3. Connect "Wire Jumpers" as follows:

Old Postion	New Position
SFB to DM5	Same
CFB to CFB*	Same (if present)
SRA to JP18, pin 2	Same (if present)

4. Set potentiometer positions visually (in reference to AG Version card) in the following positions for preliminary set-up:

P2	Same	P9	Same
P3	Factory Set	P10	Same
P4	Same	P11	Same
P5	Same	P12	Same
P8	Same		

Replacing AJ Version card with AK *or* AM Versions:

1. Transfer all terminal board components.
2. Set jumpers in the following positions (in reference to AJ Version card):

JP1	Same	JP16	Same
JP4	Same	JP17	Same
JP11	Same	JP18	Same
JP12	Same	JP19	Same
JP13	Same	JP20	Same
JP14	Same	JP21	Same

3. Connect "Wire Jumpers" as follows:

Old Postion	New Position
SFB to DM5	Same
CFB to CFB*	Same (if present)
SRA to JP18, pin 2	Same (if present)

4. Set potentiometer positions visually (in reference to AJ Version card) in the following positions for preliminary set-up:

P2	Same	P9	Same
P3	Factory Set	P10	Same
P4	Same	P11	Same
P5	Same	P12	Same
P8	Same		

**CFB on Analog Process Interface card to CFB on Main Control Card.*



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